Task 5: Digital Forensics – The Case of the Suspicious File

Objective:

The objective of this task is to perform basic digital forensic analysis on a suspicious file by verifying its integrity, identifying its type, extracting readable information, and checking for malicious indicators.

Software/Tools Used:

** Kali Linux*
** Terminal*
** file command*
** sha256sum*
** strings*
** binwalk (optional)*

Procedure:

1. Obtained the suspicious file for analysis.
2. Generated the SHA-256 hash of the file to verify its integrity.
3. Determined the file type using the file command.
4. Extracted readable strings from the file using the strings command.
5. Examined the extracted strings for suspicious URLs, IP addresses, or hidden messages.
6. Used binwalk (if required) to inspect embedded data inside the file.
7. Documented all observations.

Commands Used:

Generate SHA-256 Hash

sha256sum suspicious_download.zip

Identify File Type

file suspicious_download.zip

Extract Readable Strings

strings suspicious_download.zip

Analyze Embedded Data (Optional)

binwalk suspicious_download.zip

Expected Result

The forensic analysis provides:

** SHA-256 hash of the file.*
** Actual file type and format.*
** Readable strings extracted from the file.*
** Any suspicious URLs, IP addresses, or embedded content.*
** Basic information useful for determining whether the file is potentially malicious.*

Conclusion

The suspicious file was successfully analyzed using basic digital forensic techniques. By verifying the file's integrity, identifying its format, and extracting readable information, valuable evidence was collected to support further security analysis and incident investigation.